

RESEARCH INTERESTS

Neutrino phenomenology and astroparticle physics; flavor evolution in solar and supernova environments; dark-matter detection and dark-sector phenomenology; physics beyond the Standard Model; statistical and machine-learning inference in physics

EDUCATION

Indian Institute of Technology Bombay

2023 – 2027

B.Tech. (Honors) in Engineering Physics, Minor in Mathematics

CPI: 9.71/10

PUBLICATIONS

Preprints

[1] B. Chauhan, **S. Sahasrabuddhe**, M. Sen, “A closer look at the LZ 248 keV event through the lens of cosmic-ray boosted dark matter,” [arXiv:2609.24982](#) (2026)

Manuscripts in preparation

- **S. Sahasrabuddhe**, S. Sahoo, M. Sen, “Impact of solar density fluctuations on neutrino propagation and detection”
- **S. Sahasrabuddhe**, M. Sen, A. Yu. Smirnov, “Refraction neutrino masses: Effects of fluctuations of the dark matter field”

RESEARCH EXPERIENCE

Self-Interactions in Ultralight Dark Matter

Aug 2026 – Present

Advisors: Prof. Alexei Smirnov (ICTP, Trieste) and Prof. Manibrata Sen (IIT Bombay)

- Developed a numerical solver for the **Gross-Pitaevskii-Poisson** equations with quartic self-interaction to simulate ultralight dark-matter halos across particle masses and interaction strengths
- Studying the impact of dark-matter self-interactions on **density-dependent refractive neutrino masses**, with a focus on their ability to prevent oscillation washout over terrestrial baselines

Cosmic-Ray Boosted Dark Matter in Xenon Detectors

Sep 2026 – Present

Advisors: Prof. Bhavesh Chauhan (BITS Pilani) and Prof. Manibrata Sen (IIT Bombay)

- Developed a semi-relativistic framework for **cosmic-ray boosted dark matter** at LUX-ZEPLIN, modeling boosted-flux production, energy-loss attenuation, and shell-model nuclear responses
- Investigated the anomalous **248 keV event at LZ** as a potential signal of spin-dependent boosted dark-matter scattering via a **light mediator** [1]
- Extending the analysis to the full set of **non-relativistic EFT operators**, deriving constraints on mediator mass, and developing a more rigorous treatment of dark-matter attenuation

Supernova Properties from Neutrino Signals with Machine Learning

May 2026 – Present

Advisor: Prof. Manibrata Sen (IIT Bombay)

- Building a **neural posterior estimation** pipeline to infer supernova progenitor properties and discrete physical scenarios directly from neutrino detector spectra, with Bayesian marginalization over uncertainties
- Integrated **SNEWPY** with a simulation-based inference framework to generate detector-specific event spectra across supernova models and progenitor parameters

Neutrino Flavor Evolution in a Fluctuating Solar Density

Apr 2025 – Apr 2026

Advisor: Prof. Manibrata Sen (IIT Bombay)

- Implemented a numerical framework for neutrino flavor evolution in **fluctuating solar matter**, treating stochastic density fluctuations as white noise to obtain modified survival probabilities
- Modeled **Kolmogorov turbulence** in the solar density profile using the randomization method, and quantified its effect on survival probabilities and solar-neutrino event rates
- Derived detector event rates and constrained the fluctuation amplitude through a chi-square analysis with an Asimov dataset for Super-Kamiokande IV and SNO, with sensitivity projections for JUNO and DUNE

Solitons in Low-Energy Nuclear Dynamics

Dec 2024 – Dec 2025

Advisor: Prof. Ameeya Bhagwat (UM-DAE Center for Excellence in Basic Sciences, Mumbai)

- Reviewed low-energy nuclear dynamics using **time-dependent nuclear density functional theory** and topological methods, studying multi-soliton interactions and phase shifts
- Modeled localized nuclear wavefunctions using topological and non-topological solitons of the nonlinear Schrodinger, KdV, and sine-Gordon equations

Quantitative Research Intern, Asset and Wealth Management

May 2026 – Jul 2026

Goldman Sachs, Summer Internship Program

- Conducted a relative-value analysis of **municipal bonds**, mining and engineering features across a fragmented market to identify the key drivers of yield, and delivered the signals through a dashboard
- Researched on **categorical boosting** model, flagging bonds that trade rich or cheap against their cohorts

HONORS AND AWARDS

Research awards

- Received **Undergraduate Research Award** for outstanding research work alongside coursework 2026
- Won **Best Poster Award for THEP** at SymPhy, annual physics symposium of IIT Bombay [POSTER ↗](#) 2026

Academic honors

- Institute Academic Awardee** for outstanding academic performance given to top **48** students 2024-26
- National Initiative for Undergraduate Science Awardee** among **72** students nationwide, by TIFR 2024
- Secured an All India Rank of **487** in JEE Advanced among over **0.2 million** candidates 2023
- KVPY Fellow** (Kishore Vaigyanik Protsahan Yojana) with AIR **154** in the KVPY-SA stream 2022
- Top 35** nationally in the Indian National Chemistry Olympiad, selected for **ICHO** Training Camp 2023
- Top 48** nationally in the Indian National Mathematics Olympiad, selected for **IMO** Training Camp 2022
- Top 1%** nationally in the Indian National Physics Olympiad, qualifying for the national stage 2023
- State Topper** in the Indian National Astronomy Olympiad, qualifying for the national stage 2023

TECHNICAL SKILLS

Programming: Python, C++, Mathematica, MATLAB, $\text{\LaTeX}$, Fortran, Bash, Git, HTML, CSS, JavaScript

Scientific computing: NumPy, SciPy, Pandas, MESA, Astropy, Photutils, SNEWPY, sbi, Qiskit, Qadence

Machine learning: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, MediaPipe

KEY COURSES

Physics: Astroparticle Physics, Quantum Field Theory, Astrophysics, Elementary Particle Physics, General Relativity, Quantum Mechanics I, II & III, Statistical Mechanics I & II, Electromagnetism, Classical Mechanics, Nonlinear Dynamics, Condensed Matter Physics, Optics, and Molecular Spectroscopy

Mathematics: Real Analysis, Complex Analysis, Abstract Algebra, Linear Algebra, Fourier Analysis, Differential Equations, Numerical Analysis

Other: Machine Learning, Cryptography, Analog and Digital Electronics, Computer Programming

COURSE PROJECTS

Shocks in Core-Collapse Supernovae [CODE ↗](#) [REPORT ↗](#) [SLIDES ↗](#)

Mar – Apr 2026

Advanced Astrophysics, Prof. Rahul Kashyap

- Built numerical hydrodynamic simulations for shock formation and propagation in core-collapse supernovae, using **Burgers' equation**, **Rankine-Hugoniot** conditions and **Sedov** self-similar solutions
- Validated blast-wave solutions in uniform and non-uniform density media, and explored the emergence of sonic critical points in steep density gradients

Mass Function of Open Star Clusters [CODE ↗](#) [REPORT ↗](#)

Oct – Nov 2025

Astrophysics, Prof. Varun Bhalerao

- Acquired and reduced multi-band (*ugri*) photometry of the open clusters Berkeley 59 and King 12 from the **GROWTH-India Telescope**, cross-calibrating magnitudes against Gaia DR3
- Constructed HR diagrams and derived the cluster mass functions through isochrone fitting

Sterile Neutrinos as Dark Matter [REPORT ↗](#) [SLIDES ↗](#)

Sep – Nov 2025

Advanced Astroparticle Physics, Prof. Manibrata Sen

- Studied the production of keV-scale sterile neutrinos in the early universe through the non-resonant **Dodelson-Widrow** and resonant **Shi-Fuller** mechanisms, with thermal and lepton-asymmetry effects
- Reviewed constraints on sterile-neutrino DM from large-scale structure, the CMB, and 3.5 keV X-ray searches

Room-Temperature Quantum Memory [REPORT ↗](#) [SLIDES ↗](#)

Mar – Apr 2026

Light–Matter Interaction, Prof. Anshuman Kumar

- Simulated a three-level ^{87}Rb Λ -system with far-detuned Raman coupling, modelling the **gradient echo memory** protocol for a scalable, non-cryogenic memory operating near 80 °C
- Achieved retrieval fidelity above 50% for unknown coherent states, beating the classical **no-cloning limit**

TEACHING AND MENTORING

Teaching Assistant · *Department of Physics, IIT Bombay*

Jan 2025 – Present

- **PH 309** Quantum Mechanics II (*Prof. Pichai Ramadevi*)
- **PH 312** Electromagnetism (*Prof. Kantimay Dasgupta*)
- **PH 221** Analog Electronics (*Prof. Pradeep Sarin*)
- **PH 110** Introduction to Classical and Quantum Mechanics (*Prof. Alok Shukla*)

Present

Jan – May 2026

Aug – Nov 2025

Jan – May 2025

Institute Student Mentor, Department Academic Mentor, and Head, EP Web Team Jun 2025 – Present
Student Mentor Program, IIT Bombay

- Mentoring first-year undergraduates through the transition from school to university, after selection through a multi-stage process of statement of purpose, peer evaluations, and interviews
- Mentoring 8 sophomores in their academic and personal growth, and helping them navigate the curriculum
- Revamped the department's [academic blog ↗](#) with a new interface, academic resources and blog posts

Project Mentor

Dec 2024 – Jul 2026

Student Association of Physics Department, Maths and Physics Club; Krittika, the Astronomy Club

- Curated 4-week courses on [Electromagnetism ↗](#) and [Quantum Mechanics ↗](#) for 40+ students
- Mentored 10 students in a reading project on neutrino physics, particle physics, and quantum field theory
- Mentored 10 students in analyzing star clusters with PHANGS and Hubble data ([code](#)) ↗, determining cluster ages, locations, and gas content, and stellar masses and spectral types from HR diagrams

OTHER PROJECTS

Pinpricks on the Vortex - Galaxy Evolution with Star Clusters [CODE ↗](#) [REPORT ↗](#)

Jun 2024

Astronomy Summer Project, Krittika, The Astronomy Club

Arduino-Powered RGB Matrix: From Shapes to Minesweeper [CODE ↗](#) [REPORT ↗](#)

Mar 2025

Digital Electronics, Prof. Pradeep Sarin

Introduction to Quantum Machine Learning [CODE ↗](#)

Dec 2024

Computational Physics Projects, Maths and Physics Club

Turbulence: From Flowing Water to Molecular Clouds [CODE ↗](#) [SLIDES ↗](#)

Sep 2024

Nonlinear Dynamics, Prof. Punit Parmananda

LEADERSHIP AND OUTREACH

Institute Maths and Physics Secretary

Apr 2024 – Mar 2026

Maths and Physics Club ↗, Institute Technical Council, IIT Bombay

- Led a 15-member team in year-round mathematics and physics events, and organized the **Summer of Science** and the **Technical Summer School**, coordinating 50+ courses for over 1500 participants
- Developed [Python tutorials ↗](#) and science-outreach material, and ran **hands-on workshops** with low-cost experiments and documentaries for **students from under-resourced schools**

Junior Design Engineer, Propulsion Subsystem

Feb 2024 – Jan 2025

IIT Bombay Rocket Team (IN-SPACe recognised high-power rocketry)

- Designed and optimized over ten solid-propellant rocket-motor configurations, fabricating candidate propellants and characterizing their burn properties through Crawford bomb measurements
- Designed a snap-fit mechanism for the motor casing and validated the design through static tests

EXTRACURRICULAR ACTIVITIES

- Participated in the **Workshop on Neutrino and Dark Matter Synergies**, IIT Bombay, and the **School on Theoretical High Energy Physics and Gravity**, PANE [CERTIFICATE ↗](#)
- Secured **2nd** place in the India Quiz at the Inter-IIT Cultural Meet 2024 in Patna
- Secured **4th** place in the India Quiz at Nihilanth 2025, the Inter IIT-IIM Quizzing Fest at IIM Lucknow
- Secured **2nd** place in the IIT Bombay Sports Quiz in consecutive years, organized by the Sports Council
- Secured **1st** place in Astromania, the annual astronomy quiz conducted by Krittika, IIT Bombay
- Earned a **recreational scuba diving certification**, with dives to depths of 12 m in the Indian Ocean
- Participated in **cricket and badminton** tournaments and completed a year-long course in hockey